

MEDIUM RISK

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Risk Score: 50.0%

Your current symptom profile falls within the moderate-risk range for Long COVID. Several symptom clusters are contributing to a meaningful burden on your health and daily functioning.

■ Patient Information

Full Name	BHOOMIKA D R	Age	21
Biological Sex	Female	Ref. No.	CS-20260422-0722
Report Date	22 April 2026, 07:22	AI Models Used	4 (Ensemble)

■ Clinical Condition Overview

Moderate-risk patients commonly experience a combination of fatigue, cognitive difficulties (brain fog), and respiratory or neurological symptoms that may fluctuate in intensity. Without appropriate management, these symptoms can persist for months. A structured recovery plan and medical guidance are important at this stage.

■ Symptom Severity Profile

Symptom	Score	Level	Clinical Note
Fatigue	5.0/10	Moderate	Persistent tiredness unrelieved by rest; may indicate post-exertional malaise.
Cough	5.0/10	Moderate	Dry or productive; monitor for worsening or blood in sputum.
Breathing	5.0/10	Moderate	Assess oxygen saturation; seek care if SpO2 < 94%.
Headache	5.0/10	Moderate	Common in Long COVID; rule out secondary causes if severe/sudden.
Sleep	5.0/10	Moderate	Disrupted sleep worsens all Long COVID symptoms; prioritise sleep hygiene.
Memory	5.0/10	Moderate	Brain fog; affects concentration, word-finding, and short-term recall.
Chest Pain	5.0/10	Moderate	Requires cardiac evaluation if persistent; do not ignore.
Smell Loss	5.0/10	Moderate	Anosmia/hyposmia; smell training may aid recovery.
Fever	5.0/10	Moderate	Low-grade fever is common; monitor for spikes above 38.5°C.
Weakness	5.0/10	Moderate	Muscle weakness may reflect deconditioning or ongoing inflammation.

Visualisation — Symptom Severity

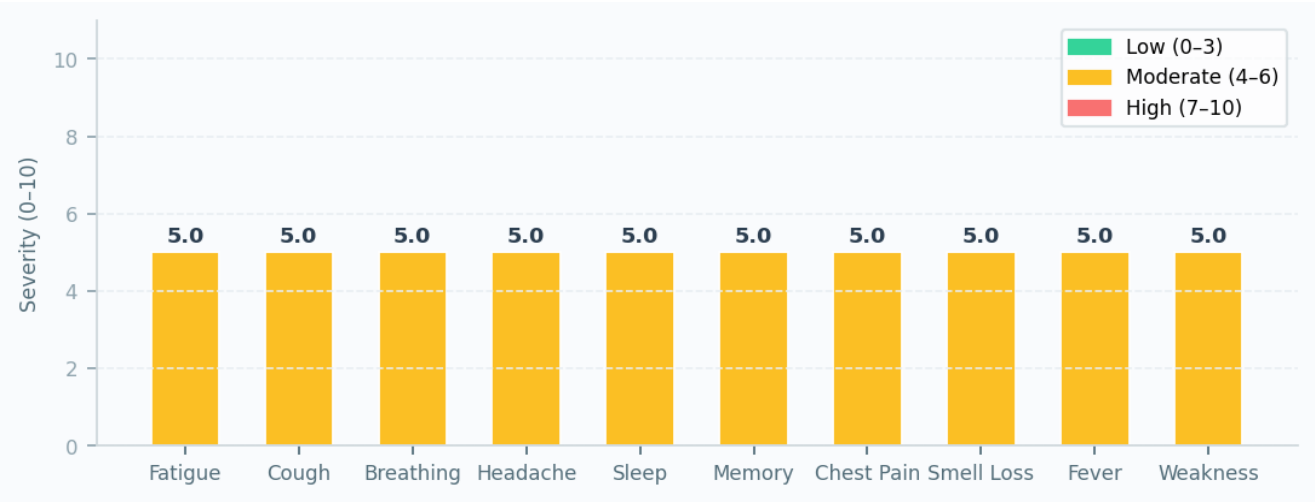
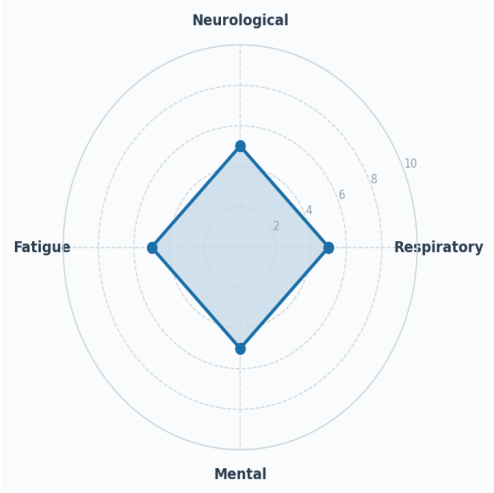


Figure 1 — Colour-coded symptom severity across all ten clinical indicators.

Category Domain Analysis



Respiratory — 5.0/10 (Moderate)

Includes cough and breathing difficulty.

Practice diaphragmatic breathing daily. Avoid cold, dry air. Use a humidifier if needed. If breathlessness at rest is present, seek urgent medical attention.

Neurological — 5.0/10 (Moderate)

Includes headache, memory/brain fog, and smell loss.

Prioritise cognitive rest alongside physical rest. Reduce screen time. Smell training (sniffing strong scents like lemon or cloves twice daily) may aid olfactory recovery.

Fatigue — 5.0/10 (Moderate)

Includes general fatigue and physical weakness.

Post-exertional malaise (PEM) is a hallmark of Long COVID. Use the pacing strategy: plan activities in small blocks, take breaks proactively, and never push through fatigue.

Mental — 5.0/10 (Moderate)

Includes sleep difficulties and mental wellbeing.

Sleep disruption worsens all other Long COVID symptoms. Maintain consistent sleep times, limit caffeine after noon, and consider a short mindfulness or breathing routine before bed.

■ AI Model Predictions

Four independent machine learning models were applied to your symptom profile. Each model predicts a cluster class: **0 = Low burden**, **1 = Moderate burden**, **2 = High burden**. Agreement across models increases diagnostic confidence.

Model	Prediction	Type	Clinical Significance
Logistic Regression	Moderate (1)	Linear discriminant	Establishes a baseline linear boundary between risk classes.
Random Forest	Moderate (1)	Ensemble tree model	Aggregates 100+ decision trees; robust to outliers and noise.
K-Means Clustering	Moderate (1)	Unsupervised clustering	Groups patients by symptom similarity; validates class membership.
Hierarchical Clustering	Low (0)	Agglomerative	Builds nested clusters; confirms population-level grouping.

■ Personalised Care Recommendations

Based on your **MEDIUM RISK** assessment and symptom profile, the following evidence-informed steps are recommended:

- 1. Consult a healthcare provider or Long COVID clinic within the next 1–2 weeks.
- 2. Practice pacing: balance activity with rest using the 'energy envelope' technique — avoid overexertion even on good days.
- 3. Consider cognitive rehabilitation exercises if brain fog is prominent (puzzles, mindfulness).
- 4. Breathing exercises (pursed-lip breathing, diaphragmatic breathing) for respiratory symptoms.
- 5. Keep a daily symptom diary to identify triggers and patterns.
- 6. Prioritise sleep hygiene: consistent bedtime, dark and cool room, no screens 1 hour before bed.
- 7. Discuss with your doctor whether physiotherapy or occupational therapy referral is appropriate.
- 8. Avoid activities that cause symptom flare-ups (post-exertional malaise is common).

■ Recommended reassessment: 2 weeks

DISCLAIMER — This report has been generated by an artificial intelligence system using self-reported symptom data. It is provided for informational and educational purposes only and does not constitute a medical diagnosis or professional medical advice. All clinical decisions must be made in consultation with a qualified and registered healthcare practitioner.